

Monomial $f(q) = a \cdot q^3$ Report

Polynomial

$$f(q) = aq^3$$

Naive

$$g_{\text{naive}}(X) = X^3 a$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 6a \frac{\Delta^2}{\epsilon} q + aq^3$$

$$\text{Var}(g_{\text{naive}}(X)) = 720a^2 \frac{\Delta^6}{\epsilon} + 324a^2 \frac{\Delta^4}{\epsilon} q^2 + 18a^2 \frac{\Delta^2}{\epsilon} q^4$$

$$\text{MSE}(g_{\text{naive}}) = 720a^2 \frac{\Delta^6}{\epsilon} + 360a^2 \frac{\Delta^4}{\epsilon} q^2 + 18a^2 \frac{\Delta^2}{\epsilon} q^4$$

$$\text{Bias}(g_{\text{naive}}) = 6a \frac{\Delta^2}{\epsilon} q$$

Unbiased

$$g_{\text{unb}}(X) = X^3 a - 6Xa \frac{\Delta^2}{\epsilon}$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^3$$

$$\text{Var}(g_{\text{unb}}(X)) = 504a^2 \frac{\Delta^6}{\epsilon} + 252a^2 \frac{\Delta^4}{\epsilon} q^2 + 18a^2 \frac{\Delta^2}{\epsilon} q^4$$

$$\text{MSE}(g_{\text{unb}}) = 504a^2 \frac{\Delta^6}{\epsilon} + 252a^2 \frac{\Delta^4}{\epsilon} q^2 + 18a^2 \frac{\Delta^2}{\epsilon} q^4$$

Comparison

$$\text{mean gap} = -6a\frac{\Delta^2}{\epsilon}q$$

$$\text{variance gap} = -216a^2\frac{\Delta^6}{\epsilon} - 72a^2\frac{\Delta^4}{\epsilon}q^2$$

$$\text{MSE gap} = -216a^2\frac{\Delta^6}{\epsilon} - 108a^2\frac{\Delta^4}{\epsilon}q^2$$

$$\text{Bias}(g_{\text{naive}})^2 = 36a^2\frac{\Delta^4}{\epsilon}q^2$$